SYSTEM VALIDATION REPORT

Electronics Store Inventory Management System (EIMS) · Team 9

Execution status	Passed	Failed	Skipped	Total
PASS	98	0	0	98

1. Purpose and Scope

This report records the system-level validation test execution for EIMS. The selected test run exercised integrated application workflows, access controls, security-related input handling, scalability checks, and transport-security configuration. The results below are transcribed from the supplied pytest terminal output.

2. Test Environment

Item	Recorded value
Operating system	Windows 11 (platform: win32)
Python	3.12.7
pytest	8.3.5
pluggy	1.7.0
pytest-html	4.2.0
Execution environment	Project virtual environment (.venv)

3. Validation Areas and Results

The command executed was: pytest tests/integration tests/security tests/test_scalability.py tests/test_transport_security.py -v --html=system_validation_report.html --self-contained-html

Validation area / test module	Passed	Failed
Integration — authentication placeholder	1	0
Integration — branch management	5	0
Integration — component image management	7	0
Integration — purchase orders	12	0
Integration — reports API	18	0
Integration — stock movements	9	0
Integration — supplier management	7	0
Integration — system availability	10	0
Security — authentication and RBAC	2	0
Security — SQL injection, input and XSS-related handling	13	0
Security — role permissions	9	0
Scalability	2	0
Transport security configuration	2	0
TOTAL	**98**	**0**

4. Key Validation Scenarios

The executed tests include branch creation and assignment rules; image upload, replacement, removal and invalid-file rejection; purchase-order creation, partial/full receipt, duplicate receipt protection and rollback; inventory search and availability reports; stock-in/out and movement history; supplier validation and permissions; monthly availability calculations; authentication and authorization checks; SQL-injection and malicious-input test cases; a 10,000-component / 20-concurrent-user scalability scenario; and TLS launcher/protocol-policy checks.

5. Execution Summary

Pytest collected 98 test cases. All 98 were reported as PASSED, with zero failures and zero skips. The observed pass rate for this selected validation run was 100% (98/98).

6. Warnings and Limitations

One DeprecationWarning was reported from Starlette's TestClient integration: the anyio.abc.BlockingPortal alias is deprecated and the warning recommends anyio.from_thread.BlockingPortal instead. It did not cause a test failure.

The scalability tests are automated tests using the project's test environment; they should not be presented as a production load-test or deployment benchmark. The transport-security tests validate TLS launcher configuration and protocol policy; they do not, by themselves, prove that a deployed production server is correctly configured. The integration authentication test is explicitly named a placeholder in the test output. This report covers only the test paths in the command above, not every test in the repository.

7. Conclusion

The selected EIMS system-validation suite completed successfully in the recorded local environment. The tested integration workflows, security checks, scalability assertions, and transport-security configuration checks all passed in this run. These results support the tested behaviours but do not replace deployment testing or a full production security assessment.